Studio North: customer economics lab

Are we moving in the right direction as a company?

Compare acquisition efficiency, contribution value and cash recovery across cohorts.

A monthly creative-work membership for independent professionals: templates save time, guided learning builds skills, and workshops add value.

24 joining cohorts: January 2024–December 2025.
Monthly renewals; cutoff January 1, 2026 (exclusive).

CRM	Renewals
Transactions	Product events
Feedback	Campaign spend
Source credit	Channel pilots

How to read the slide colors

Concept	Definitions and distinctions to use when interpreting the data.
Case application	Studio North and its connected customer records.
Evidence interpretation	Observed results, predictions, and what each can establish.
Group decision	One focused judgment, discussed with your group.

Three ideas guide the entire lab

1. Connected customer evidence

The Universal Customer View classifies each dataset's fields as Identifiers, Engagement, Relationship or Perception. It reveals what is recorded and what is missing.

2. Customer economics over time

Joining cohorts distinguish customer age from calendar time. Retention, contribution and acquisition cost determine whether relationships create value.

3. Acquisition cost and customer value

Channel holdouts estimate how many additional customers a campaign acquires. Later, compare iCAC with their assumed contribution value.

AI proposes the work. Saved SQL, data checks, held-out outcomes and your judgment make the work reviewable.

Channel holdouts: acquisition counts and iCAC

Four separate July 2025 pilots: 500 eligible prospects per arm, each followed for 31 days. Only the focal campaign is withheld from control.

Table 1. Additional acquisitions and full incremental acquisition costs

Channel / need	Buyers T / C	Additional buyers	Extra spend	Credited CAC	iCAC
Search / save time	90 / 40	50	$11,000.00	$122.22	$220.00
Search / build skill	84 / 35	49	$11,000.00	$130.95	$224.49
Partner / save time	80 / 40	40	$11,000.00	$137.50	$275.00
Partner / build skill	100 / 35	65	$11,000.00	$110.00	$169.23

Why does subtracting background acquisitions raise the cost per additional customer?

The pilots establish acquisition cost. Contribution LTV enters after the cohort model.

Every dataset gets its own Universal Customer View

Example: people.csv contains one record per operational customer. The acquisition pilots are a separate sample.

Table 2. Field classification for people.csv

Identifiers	Engagement	Relationship	Perception
person_id age_band household_type region	—	record_created_at attributes_recorded_at	primary_need

The stated need is a reported intention. Profile timestamps document when the relationship information became available.

Repeat this field-level table for all eight exports, then open the Archify map of keys, grains and connections.

Group decision 1: who needs which onboarding route?

Studio North can offer a ready-to-use template or a guided lesson. Which recorded grouping should inform these onboarding routes?

Table 3. Two candidate grouping rules

Grouping	Two recorded groups	What the field describes
Age band	18–34 / 35+	A demographic characteristic.
Stated need	Save time / Build skill	The goal stated before acquisition.

Choose one grouping and explain why it is relevant to this onboarding objective.

Seek similar customers within cells and different customers across cells. Compare retention and contribution at matched ages.

Turn the connected view into a reproducible query

Inspect raw views

Keep source columns and row meanings.

Define the customer clock

First paid month sets cohort age index 0.

Build SQL outputs

Save the exact SQL and ordered results.

Reconcile the totals

No duplicate money; no missing controls.

Figure 1. From source views to reconciled cohort evidence

Prompt to try

“Show the SQL that calculates retained customers and contribution by joining month and customer age. Explain the denominator and preserve unobserved future ages as missing.”

A SQL script makes a calculation inspectable; the data definitions determine whether it answers the right question.

A cohort follows the same original customers

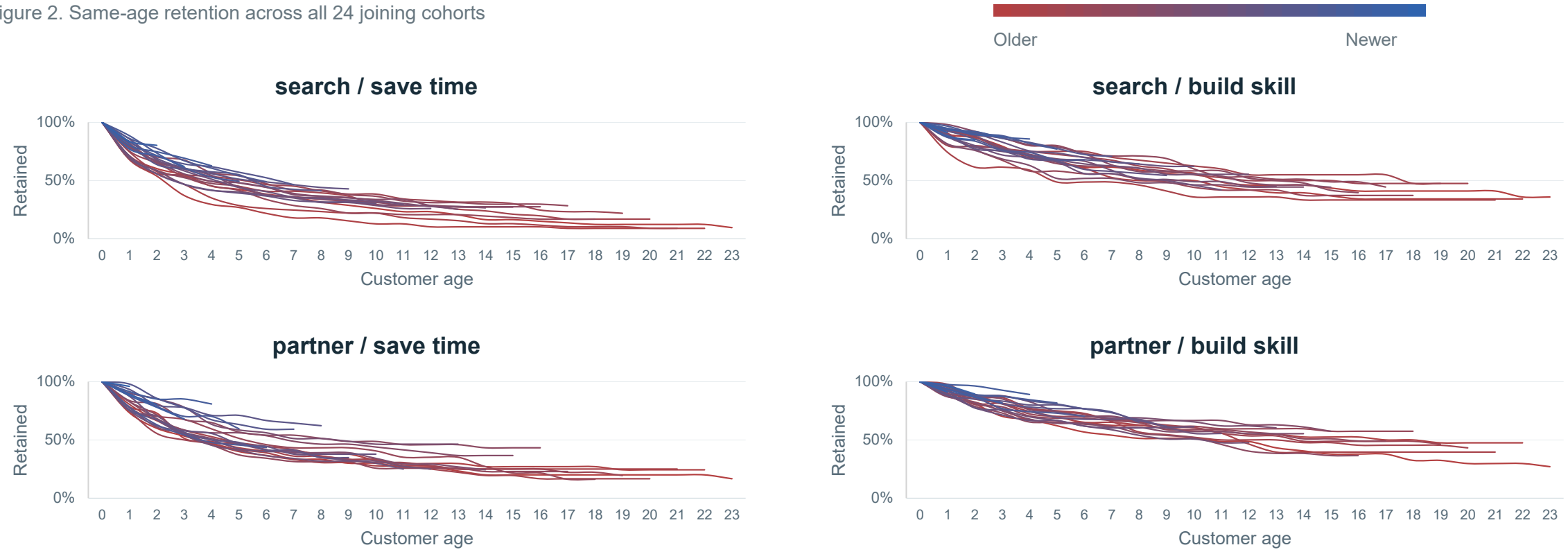
Table 4. Retained share by joining cohort and customer age (months)

Cohort	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
2024-01	100%	80%	70%	63%	55%	51%	47%	45%	42%	40%	39%	35%	34%	31%	28%	27%	26%	26%	24%	24%	23%	23%	22%	20%
2024-02	100%	78%	70%	56%	49%	46%	42%	38%	37%	36%	33%	32%	30%	29%	28%	27%	27%	26%	26%	25%	25%	25%	25%	—
2024-03	100%	78%	66%	59%	51%	44%	42%	40%	38%	36%	34%	32%	29%	27%	25%	25%	25%	24%	24%	24%	24%	24%	—	—
2024-04	100%	84%	72%	67%	62%	59%	55%	52%	49%	46%	42%	40%	38%	37%	35%	34%	32%	31%	30%	30%	29%	—	—	—
2024-05	100%	78%	67%	65%	62%	60%	57%	54%	50%	47%	45%	42%	40%	39%	38%	38%	35%	34%	34%	32%	—	—	—	—
2024-06	100%	82%	73%	65%	57%	53%	49%	46%	44%	41%	39%	37%	37%	37%	34%	32%	31%	30%	30%	—	—	—	—	—
2024-07	100%	85%	77%	72%	63%	60%	56%	53%	52%	49%	45%	42%	41%	40%	39%	38%	38%	36%	—	—	—	—	—	—
2024-08	100%	80%	72%	65%	60%	56%	52%	51%	48%	46%	45%	42%	40%	38%	37%	35%	35%	—	—	—	—	—	—	—
2024-09	100%	87%	78%	72%	66%	63%	59%	54%	50%	50%	49%	48%	43%	42%	42%	40%	—	—	—	—	—	—	—	—
2024-10	100%	82%	71%	63%	57%	53%	50%	47%	46%	44%	42%	40%	39%	38%	37%	—	—	—	—	—	—	—	—	—
2024-11	100%	87%	80%	75%	67%	60%	56%	52%	50%	48%	46%	45%	44%	42%	—	—	—	—	—	—	—	—	—	—
2024-12	100%	81%	73%	63%	57%	54%	50%	48%	44%	42%	41%	39%	38%	—	—	—	—	—	—	—	—	—	—	—
2025-01	100%	87%	73%	64%	59%	54%	49%	47%	45%	44%	41%	38%	—	—	—	—	—	—	—	—	—	—	—	—
2025-02	100%	82%	70%	64%	60%	56%	54%	50%	49%	47%	47%	—	—	—	—	—	—	—	—	—	—	—	—	—
2025-03	100%	85%	74%	69%	64%	61%	57%	55%	51%	49%	—	—	—	—	—	—	—	—	—	—	—	—	—	—
2025-04	100%	90%	78%	73%	69%	67%	63%	60%	57%	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
2025-05	100%	92%	83%	78%	73%	69%	65%	62%	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
2025-06	100%	87%	79%	70%	65%	61%	58%	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
2025-07	100%	92%	82%	76%	71%	66%	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
2025-08	100%	90%	87%	83%	78%	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
2025-09	100%	87%	81%	74%	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
2025-10	100%	89%	84%	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
2025-11	100%	89%	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
2025-12	100%	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—

Read across for one cohort over time, or down for different cohorts at the same age. Future ages stay missing.

Group decision 2: is improvement visible within groups?

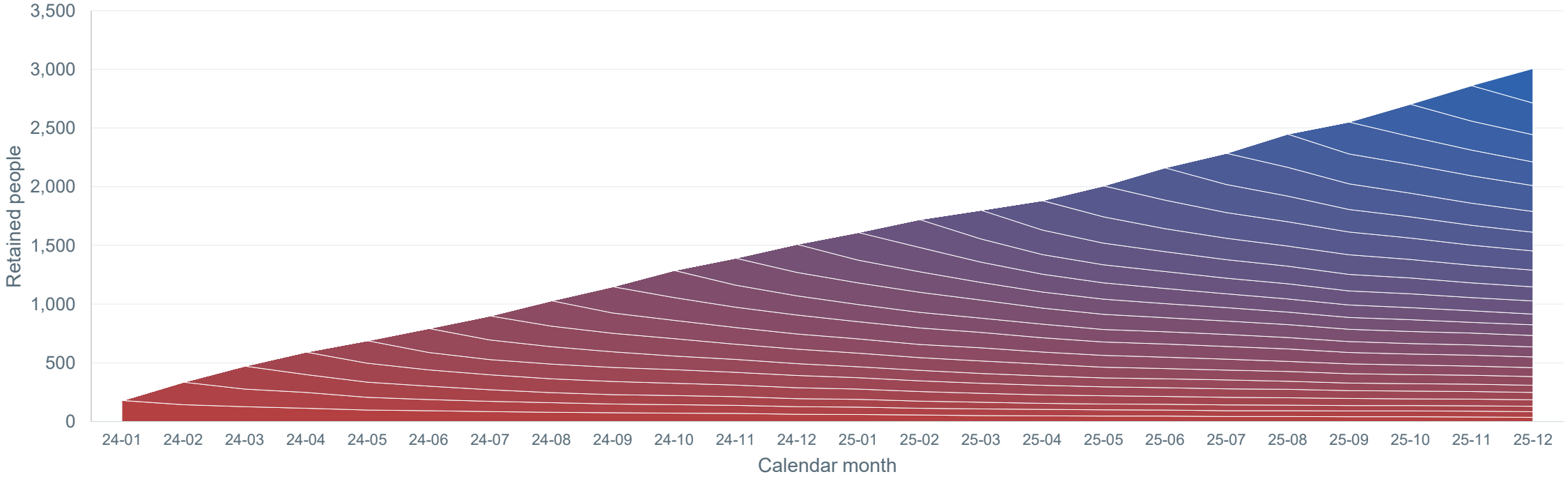
Figure 2. Same-age retention across all 24 joining cohorts



Do newer cohorts retain better at the same age? Is the progress broad or concentrated?

Joining cohorts accumulate into the company

Figure 3. Retained subscribers by joining cohort and calendar month



Each band is one joining cohort. The height is the retained customer base, not cumulative acquisitions.

Model renewal and revenue before forecasting LTV

Retention model

A hierarchical shifted Beta-Geometric model represents persistent differences in churn risk. Cohort and group information share strength where histories are short.

Revenue model

A Gamma model describes positive revenue among retained customers. Combine this forecast with retention, variable costs and observed contribution.

Validation

Compare chronological holdout predictions with a simple baseline and an Orduz-style neural network. Keep uncertainty visible; a fitted curve is not a guarantee.

Value the first 12 monthly receipts for every cohort. Net LTV subtracts CAC; cash payback tracks undiscounted recovery.

Run the locked NumPyro/JAX models on CPU with uv. Describe the model before showing its results.

Group decision 3: where should acquisition spend go?

Join each pilot's iCAC to the July 2025 channel / need cell. The model values the first 12 receipts; six receipts are observed for these cells.

Table 5. Contribution value before acquisition cost, joined to pilot iCAC

Channel / need	Forecast share	Value before CAC	iCAC	Value less iCAC
Search / save time	38%	$186.66	$220.00	-$33.34
Search / build skill	46%	$284.03	$224.49	$59.54
Partner / save time	41%	$215.07	$275.00	-$59.93
Partner / build skill	47%	$299.62	$169.23	$130.38

Which cell looks most attractive, and how sensitive is that choice to customer quality?

Planning assumption: additional buyers resemble the matched value cell. This is not measured lifetime campaign profit.

Hand your team a decision and a reproducible report

Are successive cohorts becoming more valuable relative to acquisition cost?

Table 6. The executive handoff

Report component	What your team should receive
Executive summary	The business, a supported verdict, a proposed action and what would change it.
Observed and forecast evidence	Age-aligned economics, journey diagnostics and acquisition-segment forecasts
Model and validation	Explain the model before forecasts; keep estimates and validation in the appendix
Data and reproducibility appendix	UCV tables, Archify key map and saved SQL; saved manifests accompany the report.

Bonus challenge: forecast 2026 outcomes for these 24 cohorts with another CPU-capable model. Submit for instructor-held evaluation.

Begin with a cover and contents. Present observed economics, then explain the model before its forecasts.